Operational Technology Graduate Demonstration Project Vision

Project Vision Brief

Background

To move beyond a generic software portfolio project, this project is intended to simulate the type of learning, engineering thinking and practical work an early-career contributor might undertake within an Operational Technology (OT) consulting project.

Following research into power-utility operational technology services, early-career roles, and the operational technology environment of electricity utilities, the aim is not to build a commercial SCADA or ADMS platform.

Instead, this project represents a **small fictional graduate work package within an electricity distribution system upgrade project.**

Project Objective

The objective is to demonstrate that I have proactively begun preparing for the type of work performed by an Operational Technology graduate consultant.

Rather than demonstrating only software development ability, the project should demonstrate:

- understanding of electricity distribution operations
- understanding of operational technology systems and their roles
- systems thinking
- structured engineering problem solving
- testing and validation methodology
- configuration and defect investigation
- technical communication and documentation
- practical software implementation

The project should answer the question:

“Would this person be able to become productive more quickly because they have already invested time understanding our industry and the way our projects operate?”

Fictional Project Scenario

A fictional electricity distribution utility is upgrading part of its operational control environment.

A simplified model of the distribution network has been created to support the upgrade.

As part of the upgrade project, a graduate engineer has been assigned responsibility for validating a small section of the upgraded operational workflow.

The work package includes:

- understanding the network model
- validating operational behaviour
- executing predefined test cases
- investigating unexpected behaviour
- identifying configuration or data issues
- documenting findings
- correcting the identified issue
- validating the corrected system

This mirrors the type of supervised engineering work a graduate consultant could realistically contribute to during an operational technology implementation project.

Demonstrator Scope

The demonstrator is **not** intended to replicate commercial operational technology platforms.

It is intended to simulate a very small engineering work package that demonstrates understanding of how multiple operational systems interact during a network event.

The demonstrator will include simplified representations of concepts such as:

- SCADA telemetry
- network topology
- feeder status
- alarms

- outage information
- operational workflows
- automation logic
- event logging
- engineering validation
- defect investigation

All network data will be fictional.

No real utility network or operational information will be recreated.

Core Engineering Scenario

The project will centre around a feeder fault occurring on a simplified electricity distribution network.

The validation process will demonstrate that:

- downstream sections become de-energised
- alarms are generated correctly
- affected customers are identified
- restoration is only proposed where electrically valid
- simplified feeder capacity constraints are respected
- uncertain or stale telemetry prevents unsafe actions
- every engineering action is traceable through system logs

One intentionally seeded configuration or data issue will cause a validation test to fail.

The project will demonstrate the engineering workflow required to:

1. detect the issue
2. investigate the cause
3. identify the incorrect configuration
4. implement a correction
5. repeat validation
6. confirm successful behaviour

Educational Focus

The project is equally a learning exercise and an engineering demonstration.

During development I will study and document:

- electricity fundamentals
- distribution network operation
- operational technology concepts
- SCADA
- ADMS
- OMS
- GIS
- operational workflows
- testing methodology
- engineering validation
- system integration concepts

The project documentation should clearly show how this understanding developed throughout the project.

Deliverables

The completed project will include:

- Engineering Design Brief
- Operational Technology research
- Requirements specification
- Simplified network model
- Operational workflow diagrams
- System architecture diagrams
- Data flow diagrams
- Requirements traceability matrix
- Validation test plan
- Executed validation results
- Defect investigation report
- Demonstrator application
- Technical documentation
- Final evaluation and lessons learned
- Short demonstration walkthrough

Guiding Principles

This project is **not** intended to demonstrate expertise in proprietary utility software or advanced power-system engineering.

Instead, it aims to demonstrate:

- initiative
- structured learning
- engineering thinking
- systems thinking
- attention to detail
- practical implementation
- technical communication
- an understanding of how Operational Technology consulting projects are delivered.

Every feature, document and engineering decision should be traceable back to one question:

“Does this help demonstrate the kind of understanding and contribution expected from a graduate Operational Technology consultant working on a real client project?”

Project Success Criteria

The project will be considered successful if, upon reviewing it, an experienced Operational Technology consultant could reasonably conclude:

“This candidate clearly recognised the gaps in their experience, invested time understanding our domain, and approached the project like a junior engineer contributing to an operational technology upgrade rather than simply building another software application.”
